

1. One card is drawn a pack of 52 cards, the prob. that it is a king or diamond, is :
(a) $\frac{1}{26}$ (b) $\frac{3}{26}$ (c) $\frac{4}{13}$ (d) $\frac{3}{13}$
2. The prob. of having one head in five tosses of a coin is :
(a) $\frac{5}{8}$ (b) $\frac{5}{16}$ (c) $\frac{5}{32}$ (d) $\frac{5}{64}$
3. The probability that a leap year will have 53 Sunday, is :
(a) $\frac{3}{7}$ (b) $\frac{2}{7}$ (c) $\frac{1}{7}$ (d) None
4. A speaks truth in 60% cases and B is 70% cases. The percentage number of cases in which they are likely to contradict each other in stating the same fact, if
(a) 18% (b) 36% (c) 46% (d) 23%
5. A party of 23 persons take their seats at a round table the odds against two persons sitting together is :
(a) 11 : 1 (b) 10 : 1 (c) 10 : 11 (d) None
6. Eight coins are tossed at a time the probability of getting at least 6 heads up, is :
(a) $\frac{229}{256}$ (b) $\frac{37}{256}$ (c) $\frac{57}{64}$ (d) None
7. A bag contains 4 white and 3 black balls, another contains 5 white and 3 black balls, If a bag is chosen random and a ball is drawn from it, the probability that it is white, is :
(a) $\frac{65}{112}$ (b) $\frac{67}{112}$ (c) $\frac{69}{112}$ (d) None
8. The probability of drawings a card of diamond from a pack of 52 cards, is :
(a) $\frac{1}{52}$ (b) $\frac{1}{12}$ (c) $\frac{1}{26}$ (d) $\frac{1}{4}$
9. The probability of throwing number greater than three with a die, is :
(a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{1}{6}$ (d) $\frac{2}{3}$
10. A husband and wife appear in an interview for two vacancies in the same post. The probability of husband's selection is $\frac{1}{7}$ and that of wife's selection is $\frac{1}{5}$. The probability that only one of them is selected is :
(a) $\frac{1}{5}$ (b) $\frac{2}{5}$ (c) $\frac{1}{7}$ (d) $\frac{2}{7}$
11. If A and B are any two events, then probability that exactly one out of them will occur is :
(a) $P(A) + P(B) - P(A \cap B)$
(b) $P(A) + P(B) - 2P(A \cap B)$
(c) $P(A) + P(B) - P(A \cup B)$
(d) None of these
12. If A and B are events such that $P(A \cup B) = P(A \cap B)$ then true relation is :
(a) $P(A) + P(B) = 0$
(b) $P(A) + P(B) = P(A) P\left(\frac{B}{A}\right)$
(c) $P(A) + P(B) = 2P(A) P\left(\frac{B}{A}\right)$
(d) None of these
13. A number is chosen from first 120 natural numbers. The probability that chosen number is multiple of 5 or 15 is :
(a) $\frac{1}{5}$ (b) $\frac{1}{8}$ (c) $\frac{1}{16}$ (d) None
14. An integer is chosen at random from first 200 digits. The probability that integer chosen is divisible by 6 or 8 is :
(a) $\frac{29}{100}$ (b) $\frac{1}{8}$ (c) $\frac{1}{10}$ (d) $\frac{1}{4}$
15. A and B are two events with $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$ and $P(A \cap B) = \frac{1}{5}$. The value of $P(A \cup B)$ is :
(a) $\frac{23}{59}$ (b) $\frac{23}{60}$ (c) $\frac{25}{26}$ (d) $\frac{62}{62}$

Space for Rough Work

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- 16.** The probability that a non leap year will have 53 sundays is :
(a) $\frac{1}{4}$ (b) $\frac{3}{7}$ (c) $\frac{1}{7}$ (d) $\frac{1}{6}$
- 17.** A and B are two events such that $P(A) = 0.42$, $P(B) = 0.4$ and $P(A \text{ and } B) = 0.16$. The probability of not A , i.e., $P(\text{not } A)$ is :
(a) 0.52 (b) 0.58 (c) 0.22 (d) None
- 18.** Two uniform dice marked 1 to 6 are tossed together. The probability of the total 7 in a single throw, is :
(a) $\frac{3}{56}$ (b) $\frac{1}{6}$ (c) $\frac{1}{18}$ (d) None
- 19.** A and B stand in a ring with 10 others. If arrangement of 12 persons is at random then probability that there are exactly 3 persons between A and B is :
(a) $\frac{1}{6}$ (b) $\frac{2}{11}$ (c) $\frac{3!}{12!}$ (d) $\frac{1}{4}$
- 20.** The odds against A solving a certain problem are 4 to 3 and odds in favour of B solving the same problems are 7 to 5. The probability that the problem is solved if they both try, is :
(a) $\frac{16}{21}$ (b) $\frac{5}{21}$ (c) $\frac{8}{21}$ (d) None
- 21.** For a biased die, the probability for different faces to turn up are :
Face : 1 2 3 4 5 6
Prob : .2 .22 .11 .25 .05 .17
The die is tossed and you are told that either 4 or 5 has turned up.
The prob. that is face 4, is :
(a) $\frac{5}{6}$ (b) $\frac{1}{6}$ (c) $\frac{1}{4}$ (d) None
- 22.** One die and one coin are tossed simultaneously. The probability of getting 6 on die and head on coin, is :
(a) $\frac{1}{2}$ (b) $\frac{1}{6}$ (c) $\frac{1}{12}$ (d) $\frac{3}{7}$
- 23.** Eight letters, to each of which corresponds an envelope, are placed in the envelopes at random. The prob. that all letters are not placed in right envelopes is :
(a) $\frac{5039}{5040}$ (b) $\frac{40319}{40320}$ (c) $\frac{1}{40320}$ (d) None
- 24.** Two dice are thrown, the prob. of getting odd number on one and a multiple of 3 on the other, is :
(a) $\frac{11}{36}$ (b) $\frac{25}{36}$ (c) $\frac{5}{36}$ (d) $\frac{15}{36}$
- 25.** A can hit a target 4 times in 5 shots, B 3 times in 4 shots, C twice in 3 shots. The fire a volley, then prob. that 2 shots at least hit, is :
(a) $\frac{2}{3}$ (b) $\frac{1}{6}$ (c) $\frac{3}{5}$ (d) $\frac{5}{6}$
- 26.** From a pack of 52 cards, 3 are drawn at random. The prob. that they are king, queen and knave, is :
(a) $\frac{16}{5525}$ (b) $\frac{16}{2550}$ (c) $\frac{1}{14}$ (d) $\frac{3}{13260}$
- 27.** A and B appear for an interview for two vacancies. Prob. of A 's selection is $\frac{1}{3}$ and B 's selection is $\frac{1}{5}$. The prob. that only one of them is selected, is :
(a) $\frac{2}{5}$ (b) $\frac{7}{15}$ (c) $\frac{4}{15}$ (d) $\frac{3}{5}$
- 28.** A bag contains 3 red and 3 white balls. Two balls are drawn one by one, the prob. that they are of different colours, is :
(a) $\frac{3}{10}$ (b) $\frac{3}{5}$ (c) $\frac{2}{5}$ (d) None
- 29.** In three throws with a pair of dice, the prob. of throwing doublet at least once, is :
(a) $\frac{125}{216}$ (b) $\frac{4}{9}$ (c) $\frac{91}{216}$ (d) None
- 30.** A and B take alternate turns in throwing a coin, the first to throw head wins a prize. If A has the first throw, then their chances of winning are in the ratio :
(a) 3 : 2 (b) 2 : 1 (c) 1 : 2 (d) 2 : 3

SCORE SHEET

a	b	c	d	a	b	c	d	a	b	c	d	a	b	c	d	a	b	c	d
1	☐	☐	☐	☐	7	☐	☐	☐	☐	13	☐	☐	☐	☐	19	☐	☐	☐	☐
2	☐	☐	☐	☐	8	☐	☐	☐	☐	14	☐	☐	☐	☐	20	☐	☐	☐	☐
3	☐	☐	☐	☐	9	☐	☐	☐	☐	15	☐	☐	☐	☐	21	☐	☐	☐	☐
4	☐	☐	☐	☐	10	☐	☐	☐	☐	16	☐	☐	☐	☐	22	☐	☐	☐	☐
5	☐	☐	☐	☐	11	☐	☐	☐	☐	17	☐	☐	☐	☐	23	☐	☐	☐	☐
6	☐	☐	☐	☐	12	☐	☐	☐	☐	18	☐	☐	☐	☐	24	☐	☐	☐	☐

PROBABILITY - 01

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|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (c) | 3. (b) | 4. (c) | 5. (b) | 6. (d) | 7. (b) | 8. (d) | 9. (b) | 10. (d) |
| 11. (b) | 12. (c) | 13. (a) | 14. (d) | 15. (b) | 16. (c) | 17. (b) | 18. (b) | 19. (b) | 20. (a) |
| 21. (a) | 22. (c) | 23. (b) | 24. (a) | 25. (d) | 26. (a) | 27. (a) | 28. (b) | 29. (c) | 30. (b) |

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